

PESTICIDE RESIDUES IN VEGETABLE, SOIL AND WATER SAMPLES COLLECTED FROM SOME DISTRICTS OF HARYANA

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ABSTRACT

A study was under taken during 2012-2016 to monitor the contamination, with respect to pesticide residue, of vegetable, soil and water samples collected from seven districts of Haryana. For the study, 390 samples were collected from Yamuna Nagar, Rewari, Karnal, Jhajjar, Palwal, Mewat and Narnaul districts of Haryana. Out of 390 samples, 228 vegetable, 71 soil and 91 water samples were collected to monitor pesticide residues. Chlorpyrifos, butachlor, pretilachlor, o,p-DDE, atrazine, triazophos, lambda cyhalothrin, endosulphan, p,p'-DDE, malathion, phorate, quinolphos and profenophos were commonly detected pesticides in these samples. Out of 390 samples, 102 samples (59 samples of vegetable, 31 of soil and 12 samples of water) were found contaminated with pesticides, which made respectively 25.87 per cent, 43.66 and 13 per cent of respective sample types. From the total contaminated samples, 60.78 per cent of the samples were found contaminated with chlorpyrifos; 27.45 per cent with pretilachlor and 21.57 per cent with butachlor while rest with o, p-DDE, atrazine, triazophos, lambda cyhalothrin, endosulphan, p,p-DDE, malathion, phorate, quinolphos and profenophos.

Key words: Contamination, Haryana, Pesticide residue, Soil, Vegetable, Water

India is presently the 2nd largest manufacturer of basic pesticides in Asia. It ranks 12th globally. The application of pesticides is essential in modern agricultural practices to enhance crop yield and also to control pest and disease that damage fruit and vegetables. However, it has the drawback of concomitant occurrence and persistence in vegetables, soil and water constituting a possible risk to consumers (Fenoll *et al.*, 2007). The direct result of application in field is the presence of pesticides residues remaining in the soil (Businelli *et al.*, 1992). Pesticides primarily move from agricultural field to surface waters in surface run-off. The amount lost from fields and transported to surface water depends on several factors, including soil characteristics, topography, weather, agricultural practices, chemical and environmental properties of individual pesticides which enter the soil by deposition from the plant surface during air drift or by washing off from the plant surface during rainfall or irrigation and then get mixed (Richards RP *et al.*, 1993; Leonardo *et al.*, 1990). In addition, pesticide residues constitute a danger to soil micro fauna and micro floral. Their toxic effects manifest on human beings when bioaccumulation occurs along the food chain after initial plant uptake. The quantity of pesticides absorbed by a given plant generally depends upon the solubility in water, quantity within the soil and the organic matter content of soil. Government and international organisations (FAO and WHO) have fixed maximum residue level (MRLs) limits for the pesticide residues in imported and locally grown

agricultural products (Pal *et al.*, 2006; Dogenheim *et al.*, 1996). MRL is a product limit and is based on the application of pesticides on crops according to good agricultural practices in controlled field experiments. It represents the maximum concentration of that residue (expressed in milligram per kg). Exceeded MRL values are strong indicator of violation of good agricultural practices (Chen *et al.*, 2011). The average pesticide consumption in India is a low i.e. 28 g per ha as compared to global average of 900g per ha (Angihotri *et al.*, 2000).

Monitoring of pesticides residues in water, vegetables and soil for an extensive evaluation of their quality is a priority objective of pesticide research to avoid possible risk to human health. To determine the residue of these chemicals in water, soil and vegetables or in a variety of food matrices, many highly sensitive and selective analytical procedures have been developed. GC-MS has been used successfully for the analysis and confirmation of many volatile pesticides.

Haryana is primarily an agricultural state. About 70 percent of residents are engaged in agriculture. Haryana is one among large contributors to India's central pool of food grains. About 86 percent of the area is arable, and of that 96 percent is cultivated. About 75 percent of area is irrigated through tubewells and an extensive system of canals. Currently in Haryana, the most important technology for ground water irrigation is the use of tubewells with pump.

Keeping this in view, this study was planned to assess pesticide residue levels in vegetable, soil and water samples.

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MATERIALS AND METHODS

Chemicals and reagents

Certified reference materials (CRM) for the study were procured from Sigma-Aldrich and Accustandard Inc. (USA). The solvents and other reagents for the extraction were procured from Merck Chromatographic grade. Anhydrous sodium sulphate, magnesium sulphate (both for drying of extract moisture), sodium chloride, acetonitrile (ACN), methanol and n-hexane (HPLC grade) were used for extraction.

Preparation of standard and stock solution

The standard stock solution for each pesticide was prepared from the certified reference material (Sigma-Aldrich) of pesticide having specific purity with uncertainty value mentioned on it. Approximately, 100 ppm stock solution was prepared using HPLC grade acetone and n-hexane. Further dilution was done to lower concentration by using n-hexane for making working standard as per requirement of analysis. Standard stock solution and working standards were stored in laboratory refrigerator at 4°C.

Sampling

Approximately two kg of vegetables samples were taken from market and field, similarly one kg of soil sample from agricultural field and one liter of water from irrigation sources like pond, tubewell and canal were collected.

Sample preparation, extraction and analysis

Extraction of vegetables

QuEChERS sample preparation method for pesticides was applied to all vegetable samples (Anastassiades *et al.*, 2003).

Samples were ground homogeneously (approx. 2 kg)

Ground samples were weighed at 10 g ± 0.1 g

Then sample/matrix was extracted using 10 ml acetonitrile (HPLC grade solvent) in 50 ml centrifuge tube

Four grams of MgSO₄ and 1 g NaCl were added, vortexed on rotospin for 10-15 minutes and centrifuged for 5 minutes at 4000-5000 rpm

1.5 ml aliquot was collected from supernatant layer into 2 ml microcentrifuge tube

For clean-up of extractants, PSA 37 mg, MgSO₄ 125 mg and C-18 37 mg of varying proportion were added and shaken for 1 minute on vortex and put in centrifuge for 3 min at 4500rpm

↓
1 ml of aliquot from supernatant layer was taken in glass tube and evaporated to near dryness
↓
0.5 ml of n-hexane was added to glass tube and stirred
↓
Transferred the solution to GC vials for confirmation and quantification of analytes by GC-MS
Extraction of soil samples
Soil samples were mixed homogeneously (Mishra *et al.*, 2017)
↓
Weighed 20 g in 250 ml stoppered flat bottom round flask
↓
extracted in 50 ml MeOH (HPLC grade)
↓
Shaken for 1 hr on auto-shaker at 160 rpm and then kept it undisturbed on flat surface for half an hour
↓
20 ml of supernatant was pipetted out and passed through anhydrous sodium sulphate (Na₂SO₄) bed into 50 ml flat bottom round flask
↓
Finally washed the anhydrous sodium sulphate bed by 10 ml of methanol into same flat bottom round flask
↓
Solvent evaporated near to dryness by rotary evaporator under vacuum (500 mm of Hg) in water bath at constant temperature 60°C
↓
Final volume was reconstituted with n-Hexane
↓
Reconstituted sample was prepared for GC-MS analysis
Extraction of water samples
Took 500 ml of water sample in a 1000 ml separating funnel (Mishra S *et al.*, 2016)
↓
Added 100 ml saturated (NaCl) brine solution to produce salt out effect
↓
Extracted by shaking well for 4-5 minutes with Dichloromethane and Hexane (80:20) thrice (100 ml, 50 ml, 50 ml)
↓
Collected the DCM (lower) organic layer extract in flat bottom flask by passing through anhydrous sodium sulphate for absorbing moisture
↓
Collected organic phase and dried in a vacuum rotary evaporator
↓
Finally reconstituted with 5 ml n-Hexane (HPLC grade) for analysis in GC-MS

Instrumentation

For residue analysis, Agilent Technology (7890A) gas chromatography system with mass (model (5975C inert XL EI/CI MSD) triple axis detector was used. Residues were separated through DB-5MS fused silica capillary column (Agilent J&W GC column, 5 % Phenylated methyl siloxane, 30 m length $\times$ 0.25 mm i.d. $\times$ 0.25 μ m film thickness) was used with linear flow at 1 ml/min. One μ l of extract was injected in splitless mode with carrier gas helium (grade-1). The pesticides were separated with 40.75 min oven programming of initial temperature of 50^o C for 2 min followed by a ramp rate of 8^oC /min up to a temperature of 280^oC temperature with hold time 10 min. The injector was operated in splitless mode at 280^oC temperature. The interface temperature was 280^oC, ion source at 250^oC and quadruple temperature at 150^oC. The MS was operated in Electron Impact Mode (EI) with electron energy 70ev, using selective ion monitoring (SIM) mode with solvent delay time 3min. This mode offer enhances selectivity over either full scan or selected ion monitoring (SIM). The presence of pesticide residues in collected vegetables, soil and water samples were quantified by comparing the retention time, peak area and peak height of the sample with those of the standards.

RESULTS AND DISCUSSION

The present study was undertaken to evaluate the pesticide residues from seven districts of Haryana region. Contamination of pesticide in collected vegetable, soil and water samples was identified by comparing their retention time, molecular weight, qualifier ions (m/z) with respect to their certified reference standards in GC-MS.

In the present research work, an attempt has been made to estimate multi-residues via developed and published scientific methods and the results indicated that the collected samples were contaminated with various pesticides commonly used in agriculture such as organochlorine, organophosphorous, herbicides and synthetic pyrethroides.

Sample collection site, no. of samples and detected pesticides are given in Table 2. From the total vegetable, soil and water samples collected from different districts, 102 samples were contaminated (vegetables 59, soil 31, and water 12) were detected with pesticides. All the samples were collected from Yamuna Nagar, Rewari, Karnal, Jhajjar, Palwal, Mewat and Narnaul during the periods of 2012-2016. Samples were found contaminated with different pesticides residues such as chlorpyriphos, butachlor, pretilachlor, o,p-DDE, atrazine, triazophos, lambda cyhalothrin, endosulphan, p,p'-DDE, malathion, phorate, quinolphos and profenophos. Among all the contaminated samples, 62 samples were found contaminated with chlorpyriphos with concentration range of 0.01-3.63 mg/kg, 22 samples with butachlor at concentration range 0.0017-1.88 mg/kg, 28 samples with pretilachlor at concentration range 0.001-1.78 mg/kg, nine samples with o,p-DDE at concentration range of 0.014-0.04 mg/kg, 5 samples with atrazine at concentration range of 0.026-2.23 mg/kg. Likewise, three samples were found contaminated with triazophos with concentration range of 0.01-2.31 mg/kg, three samples with lambda-cyhalothrin at concentration range of 0.73-4.22 mg/kg and two samples with cypermethrin with concentration range of 0.313-3.37 mg/kg. p,p'-DDE, endosulphan were detected in 2 samples each with concentration range of 0.016-0.42 and 0.03-0.55

Table 1. Pesticide concentration ranges found in the tested samples

Pesticide	Positive sample (No.)	Detection (%)	Range (mg/kg)	MRL value (μ g/lit)			Sample exceeding MRL (No.)		
				Veg.	Soil	Water	Veg.	Soil	Water
Chlorpyriphos	62	57.94	0.001-3.63	0.2	0.03	0.3	33	19	2
Pretilachlor	28	26.16	0.001-1.78	-	-	-	9	4	0
Butachlor	22	20.56	0.017-1.88	-	-	-	6	2	0
OP-DDE	9	8.41	0.014-0.04	3.5	0.05	-	-	0	0
Atrazine	5	4.6	0.026-2.23	0.2	0.05	-	-	3	0
Triazophos	3	2.80	0.01-2.31	0.2	-	-	2	0	0
L-Cyhalothrin	3	2.80	0.73-4.22	-	0.01	-	2	1	0
Cypermethrin	2	1.86	0.31-3.37	0.20	-	-	2	0	0
PP-DDE	2	1.86	0.016-0.42	3.5	0.05	-	1	0	0
Endosulphan	2	1.86	0.03-0.55	2.0	0.04	-	0	1	0
Quinolphos	1	0.93	0.060	0.2	-	-	1	0	0
Profenophos	1	0.93	0.58	-	-	-	1	0	0
Malathion	1	0.93	2.42	3.0	0.05	0.5	-	1	0
Phorate	1	0.93	1.59	0.05	0.05	-	1	0	0

Table 2. Pesticide contamination in vegetable samples collected from different districts in Haryana

District	Total sample size	Positive	Above (MRL) (µg/lit)	Samples with multi-residue	Pesticide detected
Yamunanagar	16	4	4	1	Chlorpyrifos, butachlor
Rewari	16	5	5	0	Chlorpyrifos, butachlor
Karnal	115	8	7	0	Chlorpyrifos, butachlor, quinalphos, profenophos, B-endosulfan, PP-DDE, L-cyhalothrin, pretilachlor
Jhajjar	20	8	8	2	Chlorpyrifos, butachlor, lambda cyhalothrin, cypermethrin, atrazine
Palwal	16	14	14	1	Chlorpyrifos, pretilachlor
Mewat	21	13	13	4	Chlorpyrifos, pretilachlor, butachlor, triazophos
Narnaul	24	7	7	4	Chlorpyrifos, pretilachlor, phorate, triazophos, butachlor
Total	228	59	58	12	

mg/kg. Quinolphos, profenophos, malathion and phorate were detected in 4 samples with concentration range 0.060, 0.58, 2.42 and 1.59 mg/kg respectively. Frequencies and pesticide concentration ranges are given in Table 1.

Among the vegetable samples, 16 samples were collected from Yamunanagar in which 4 samples were found pesticide contaminated with chlorpyrifos, butachlor and all samples were found above MRL. Sixteen samples were collected from Rewari districts in which 5 samples were found contaminated with chlorpyrifos, butachlor and all positive samples were found above MRL. From Karnal district, 115 samples were collected in which 8 samples were found positive for chlorpyrifos, butachlor, quinalphos, profenophos, B-endosulphon, PP-DDE, L-cyhalothrin, pretilachlor and 7 samples were found above the MRL. Twenty samples were collected from Jhajjar, out of which 8 samples were found positive with chlorpyrifos, butachlor, lambda-cyhalothrin, cypermethrin, atrazine and all the samples were found above

the MRL. Further, 16 samples were collected from Palwal in which 14 samples were found positively contaminated with chlorpyrifos, pretilachlor and all samples were found above the MRL. Likewise, 21 samples were collected from Mewat in which 13 samples were found positively contaminated with chlorpyrifos, pretilachlor, butachlor, triazophos and all samples were found above the MRL. 24 samples were collected from Narnaul in which 7 samples were found positively contaminated with chlorpyrifos, pretilachlor, phorate, triazophos, butachlor and all samples were found above the MRL. MRL values were taken from Handbook for Pesticides Residue Manual, Indian Council of Agricultural Research, June-2007. Samples taken from Palwal and Mewat were highly contaminated whereas Karnal samples were least contaminated. It clearly indicates that use of organophosphorus pesticides like chlorpyrifos, quinalphos, triazophos, phorate, and herbicides like pretilachlor, butachlor are common in vegetables of the area.

Table 3. Pesticide contamination in soil samples collected from different districts in Haryana

District	Samples collected	Positive	Above (MRL) (µg/lit)	Multi-residue samples	Pesticide detected
Yamuna Nagar	3	2	2	0	Chlorpyrifos
Rewari	3	0	-	0	-
Karnal	45	20	20	14	OP-DDE, butachlor, chlorpyrifos, atrazine, PP-DDE, endosulfan, L-cyhalothrin, triazophos, pretilachlor, cypermethrin
Jhajjar	5	5	5	0	Chlorpyrifos, atrazine, malathion
Palwal	5	2	2	0	Chlorpyrifos
Mewat	5	1	1	0	Chlorpyrifos
Narnaul	5	1	1	0	Chlorpyrifos
Total	71	31	31	14	

Table 4. Pesticide contamination in water samples collected from different districts in Haryana

District	Samples drawn	Positive	Samples above (MRL) (µg/lit)	Samples with multi-residue	Pesticide detected
Yamunanagar	5	0	-	-	-
Rewari	6	0	-	-	-
Karnal	38	10	0	-	Pretilachlor, butachlor
Jhajjar	10	-	-	-	-
Palwal	6	-	-	-	-
Mewat	10	1	1	-	Chlorpyrifos
Narnaul	16	1	1	-	Chlorpyrifos
Total	91	12	12	-	-

From a total of 71 soil samples, 31 samples were found contaminated with pesticide residue as shown in Table 3. Three samples were collected from Yamunanagar in which 2 samples were found contaminated with chlorpyrifos, and both samples were found above MRL. All the three samples collected from Rewari district samples were found below the detection limit. Among 45 samples collected from Karnal, 20 samples were found contaminated with OP-DDE, butachlor, chlorpyrifos, atrazine, PP-DDE, endosulfan, L-cyhalothrin, triazophos, pretilachlor, and all samples were found above the MRL. Five samples were collected from Jhajjar which were found contaminated with chlorpyrifos, atrazine, malathion and all the samples were found above the MRL. Further, five samples were collected from Palwal in which two samples were found contaminated with chlorpyrifos and all samples were found above the MRL. Likewise, among five samples collected from Mewat one was found positively contaminated with chlorpyrifos (above MRL). Out of five samples collected from Narnaul district, one was found positively contaminated with chlorpyrifos and which was also found above the MRL.

From a total of 91 water samples, 12 samples were found contaminated with pesticide residue as shown in Table 4. Five samples were collected from Yamunanagar, 6 samples from Rewari district, 6 samples from Palwal and 10 samples were collected from Jhajjar, which were found below the detection limit. Thirty eight samples were collected from Karnal in which 10 samples were found positively contaminated with butachlor, pretilachlor, and all samples were found above the MRL. Likewise, 10 samples were collected from Mewat in which single sample was found positively contaminated with chlorpyrifos which was above the MRL. Sixteen samples were collected from Narnaul in which one sample was found positively contaminated with chlorpyrifos which was also found above the MRL. Water samples were found contaminated commonly with chlorpyrifos, pretilachlor, butachlor. Water samples were not as contaminated as soil and vegetable samples. Ten samples out of 38 from Karnal and one each out of 10 and 16 samples from Mewat and Narnaul, respectively, were found contaminated with pesticide residues.

Overall, vegetables were found more toxic with pesticides as compared to soil and water. Chlorpyrifos, butachlor, pretilachlor are common pesticides present in soil, water and vegetables. Chlorpyrifos is the most commonly used pesticide of the area. From total 59 contaminated samples of vegetables, 58 samples were found above MRL and 12 samples were found contaminated with multi-residue, whereas from 31 contaminated soil samples, all were above MRL and 14 out of them were contaminated with multi-residue. Similarly, 12 water samples were found contaminated in which 2 samples were above MRL.

To conclude, study of persistence of pesticide residue in Yamuna nagar, Rewari, Karnal, Jhajjar, Palwal, Mewat and Narnaul districts of Haryana state of India revealed that pesticides were found in about 26 % of the samples collected during 2012-2016. Among the contaminated samples, 25.87% vegetables, 43.66 % water and 13 % water samples were found contaminated. From the total contaminated samples, 60.78 % of the samples were found contaminated with chlorpyrifos; 27.45 % with pretilachlor and 21.57 % with butachlor whereas rest were contaminated with OP-DDE, atrazine, triazophos, lambda cyhalothrin, endosulphan, PP-DDE, malathion, phorate, quinolphos and profenophos. This study will inform farmers, consumers, researchers and policy makers.

Authors' contribution

Conceptualization and designing of the research work (LKT, SA, BSS); Execution of field/lab experiments and data collection (MKS, MKS, SM); Analysis of data and interpretation (MKS, MKS, SM); Preparation of manuscript (SM, V)

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